

Adventure Works Sales Analysis

Detailed Project Summary

Overview

Adventure Works Sales Analysis is an Excel-based sales analytics project built to examine four years of transactional data and convert it into meaningful business insights. The project was designed to understand sales and profit performance across time, products, customers, geography, and demographic segments through a structured data model and interactive dashboards.

Data Preparation and Modeling

The raw dataset was cleaned and transformed using Power Query to make it suitable for analysis. A data model was then built using six connected tables: Fact Internet Sales, Dim Sales Territory, Dim Product, Dim Geography, Dim Date, and Dim Customer. Relationships were created between these tables to enable flexible and dynamic reporting. Measures were created to calculate key business metrics, and M code was used to derive age for deeper customer analysis. A macro was also applied to make dashboard navigation easier by clearing filters efficiently.

Dashboard Structure

The workbook includes two main dashboards, each designed to answer different business questions and provide a complete view of performance.

Dashboard 1: Time Analysis

- KPI comparison with the previous year.
- Yearly performance metrics.
- Monthly profit trends.
- Profit by week type.
- Quarterly profit analysis.
- Profit by weekday.

This dashboard helps identify seasonality, trend shifts, and periods of strong or weak performance.

Dashboard 2: Product and Customer Analysis

- Top 5 profitable products.
- Top 5 profitable customers.
- Profit by gender.
- Profit by product color.
- Profit by pricing type.
- Country-wise profit.
- Profit by age group.

This dashboard helps identify which segments contribute most to overall business success and where profitability is concentrated.

Key Insights

- Profitability is concentrated in a small number of products and customers.
- Performance changes across months, quarters, and weekdays, showing clear time-based patterns.
- Geographic variation suggests that some countries contribute far more profit than others.
- Product attributes such as color and pricing type influence profit levels.
- Age and gender provide useful segmentation for targeted business decisions.

Business Value

The project demonstrates how raw transactional data can be converted into a clear business story that supports decision-making. By combining data cleaning, modeling, KPI design, and dashboard visualization, the analysis reveals practical opportunities for improving product strategy, customer targeting, geographic focus, and time-based planning.

Skills Demonstrated

- Data cleaning and transformation.
- Data modeling in Excel.
- Power Query and Power Pivot.
- KPI and measure creation.
- Dashboard design and interactivity.
- Business insight generation.
- Analytical storytelling.

Conclusion

Adventure Works Sales Analysis is a strong example of Excel-based business intelligence in action. It shows the full workflow from raw data to structured model to interactive dashboards, and it highlights the ability to extract actionable insights from transactional sales data.